Workbook: Bone Histology

Week 2 workbook · BIO 004 Human Anatomy · Summer 2026

A comprehensive workbook on cartilage and bone at the tissue level: the three cartilage types and how cartilage grows, bones classified by shape, the gross anatomy of a long bone, compact and spongy bone microanatomy, the four bone cells, how bone forms (intramembranous and endochondral ossification), and the zones of the growth plate. Use it to check your understanding before exam questions push you on labels.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Cartilage as a tissue · 5 questions

Q01 · DOK 1 · Recall

List the general features of cartilage as a tissue: cells, matrix, vascularity, and innervation.

Q02 · DOK 2 · Compare

Compare the three cartilage types by distinguishing feature and one typical location.

Q03 · DOK 1 · Recall

What is the perichondrium, what is it made of, and what does it do for the cartilage it covers?

Q04 · DOK 2 · Compare
Compare appositional and interstitial growth of cartilage.

Q05 · DOK 1 · Recall
Distinguish chondroblasts from chondrocytes.

Bone classification · 5 questions

Q06 · DOK 1 · Recall
Name the five categories of bones by shape and give one example of each.

Q07 · DOK 1 · Recall
Name the major parts of a long bone in order from one end to the other.

Q08 · DOK 1 · Recall
Distinguish the epiphyseal plate from the epiphyseal line.

Q09 · DOK 1 · Recall
Describe the periosteum and what it does.

Q10 · DOK 1 · Recall
What is the endosteum and where does it sit?

Bone microanatomy · 7 questions

Q11 · DOK 1 · Recall
Describe the osteon, the structural unit of compact bone, including its named parts.

Q12 · DOK 2 · Compare
Compare Haversian and Volkmann (perforating) canals.

Q13 · DOK 1 · Recall
What are lamellae in bone, and what is the difference between concentric, interstitial, and circumferential lamellae?

Q14 · DOK 1 · Recall

What are trabeculae, where do you find them, and how are they organized?

Q15 · DOK 2 · Compare

Compare compact and spongy bone by structure, location within a bone, and functional role.

Q16 · DOK 1 · Recall

Name the four bone cells and what each one does.

Q17 · DOK 1 · Recall

Osteoclasts have an unusual origin compared with the other bone cells. What is it?

Bone formation and growth · 5 questions

Q18 · DOK 2 · Compare

Compare intramembranous and endochondral ossification by what each starts from and which bones each forms.

Q19 · DOK 2 · Apply

In broad strokes, what happens during endochondral ossification of a long bone?

Q20 · DOK 2 · Compare

How does a long bone grow in length, and how does it grow in width?

Q21 · DOK 2 · Apply

Name the zones of the epiphyseal plate from the epiphysis toward the diaphysis, and state what is happening in each.

Q22 · DOK 2 · Apply

At the end of long bone growth, what happens to the epiphyseal plate, and how does this show on an X-ray?

Clinical reasoning and skeletal imaging · 4 questions

Q23 · DOK 3 · Reason

A child's long bone fractures through the epiphyseal plate. Predict why this injury is more concerning than a similar break in an adult bone.

Q24 · DOK 2 · Apply

On an X-ray of a child's knee, you can see a dark band of cartilage in each long bone near the joint. What is that, and what would it look like in an adult X-ray?

Q25 · DOK 3 · Reason

Osteoporosis is a disease in which osteoclast activity outpaces osteoblast activity. Using what you know about bone remodeling, predict the structural consequences in compact bone and in spongy bone.

Q26 · DOK 2 · Apply

Vitamin D is needed to absorb calcium from the gut. Using what you know about bone matrix, predict what happens to a bone when vitamin D is severely deficient in a growing child versus an adult.